

Mth621

*Appreciation of e

*Square root of 8 is irrational

*Alternative series def + example

* 2 question for proof

Past papers sy Mcqs ahy tha

9:41 AM

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Math621

- 1.define power series
 - 2.convergence test ka question tha
 - 3.prove $n^{\frac{1}{n}} \rightarrow 1$
 - 4.monotonic sequence ka theorem tha
 5. cauchy convergence criterion ka theorem tha
- Ik or theorem tha , baqi. 7 MCqs ahmad file sy thy

12:29 P

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MTH621 paper

Q1: State completeness property of $\mathbb{R}$

Q2: Cauchy convergence criterion

Q3: convergence test ka 1 question

Q4: open cover of $[0,1]$

Q5: proof tha chota sara sequence k topic se

Q6: Rabe test ka question tha

.

MCQs sary Ahmad ki file se thy, mtlb wo 20 k 20 mery theek hen 🙏

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MTH 621 MID Term Paper 30/12/2018

Q No 1) If $x \in \mathbb{R}$, $y \in \mathbb{R}$ and $x < y$ then there exist $p \in \mathbb{Q}$ such that $x < p < y$. (5 marks)

Q No 2)

$$\sum a_n = \sum \frac{n!}{\alpha(\alpha+1)\dots(\alpha+n-1)}, \alpha > 0 \rightarrow (5 \text{ marks})$$

Check Series Convergence or not by Raabe's test.

Q No 3) Define Ratio test. (3 marks)

Q No 4) $\sum \frac{x^n}{n!}$ Find Radius of Convergence. (3 marks)

Q No 5) $\sum \frac{x^n}{n!}$ Sequence $(n - n^2)$ check Cgs or dgs ~~by B.T.~~ (2 marks)

Q No 6) Let $S = (-\infty, -1] \cup (1, 2) \cup \{3\}$
Then the set of limit points of S is $(-\infty, -1] \cup [1, 2]$. (2 marks)

Good Luck

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MTH 621 paper

Questions:

- 1- Suppose that f is differentiable on $[a, b]$, $f'(a) \neq f'(b)$ and y is b/w $f'(a)$ and $f'(b)$. Then $f'(c) = y$ for some c in (a, b)

Prove: kanna tha

5-Marks

- 2- prove Generalize Mean Value thrm

5-Marks

- 3- If f is continuous on $[a, b]$ then f is integrable on $[a, b]$

- 4- Define Upper Integral?

- 5- Determine f is integrable on $\frac{1}{1+x}$

- 6- If f is differentiable at x_0 then

f is continuous

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1- Discuss the continuity of

$$\begin{cases} x+1 & \text{if } x \geq 0 \\ x-1 & \text{if } x \leq 0 \end{cases} \quad \text{at } x=0$$

2- Define Dirichlet's Test

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$$\sum_{n=2}^{\infty} \frac{\sin n\theta}{n + (-1)^n}$$

$$\theta \neq 1 < n.$$

Q14

Is

$$\left(\frac{1}{n} + n^2 \right)$$

bounded?

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(22)

ABP's Test

Series $\sum a_n b_n$ Converges.
if $a_{n+1} \leq a_n$ For $n \geq k$
 $\lim_{n \rightarrow \infty} a_n = 0$ and

$$|b_k + b_{k+1} + \dots + b_n| \leq M, n \geq k$$

Q#1:- Cauchy convergence criterion - (5)

Q#2:- $\sqrt{8}$ is always irrational number (5)

Q#3:- Show that $(n - n^2)$ converges or diverges. (2)

Q#3:- Give an example of subsequence which does not converges but its subsequence does converges: (3)

Q#4:- $S \supseteq (-\infty, -1) \cup (1, 2) \cup (3)$ give all limit points of S . (3)

Q#5:- $\{S_n\} = \left\{ \frac{1}{n} \right\} = \left\{ 1 + \frac{1}{2}, \frac{1}{3}, \dots \right\}$
write subsequence :- (2)

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Mth621 my today's paper

1) state convergence sequence

2) is irrational is countable

3) check convergence of series

Long

4) $\sqrt{8}$ is irrational prove

5) prove "e"

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Mth621

Mcqs conceptual th

Infimum definition

Series power

Bounded sequence thrm tha...

Underroot 8 rational hai yh prove
krna tha 5 marks

RABEES TEST 5 marks

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Write any subsequence of sequence $1/n^2$
2marks

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Define infimum 2marks

Discuss series convergent or divergent $(n/2 + 1/n)$
3marks

Cauchy's criterion state and proof 5

Aik theorem tha koi yad nai wo b 5 ka
tha

12:34 PM

MTH 621

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Mcqs were v-tough and
only two mcqs from quiz.

- 1) Sequence Given thi uski Limit
find krni thi- (2)
- 2) Power Series def. (2)
- 3) Prove that Convergent Series
is bounded. (3)
- 4) Define Limit inferior for S_n - (3)
- 5) Find the approximation of e . (5)
- 6) Principle of mathematical
induction se 1 question- (5)

Question # 1 of 10 (Start time: 08:10:51 PM, 07 November 2017)

Which one is not an initial segment of $\mathbb{N}$

Select the correct option

☐	$\{1, 2, 3, \dots, 30\}.$	<i>AHMAD M-SC(MATHS)</i> <i>GOLD MEDALIST</i> <i>areeb2113@gmail-com</i>
☐	$\{1, 2, 3, \dots, 100, 101, 102, 103\}.$	
☐	$\{1, 2, 3\}.$	
☒	$\{2, 4, 6\}.$	

The supremum of the set $B = \{p \in \mathbb{Q} : p^2 < 3\}$ is

▶ Select the correct option

☐	$\sqrt{2}$. <i>AHMAD M-SC(MATHS)</i> <i>GOLD MEDALIST</i> <i>areeb2113@gmail.com</i>
☐	$\sqrt{3}$.
☐	10.
☐	19.

Question # 3 of 10 (Start time: 08:12:24 PM, 07 November 2017)

The union of set of rational numbers and the set of irrational numbers is

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Select the correct option



The set $\{n \in \mathbb{Q} : x^7 - 10x + 30 = 0\}$.



The set of integers.



The set of natural numbers.



$\mathbb{R}$, the set of real numbers.

Question # 4 of 10 (Start time: 08:13:00 PM, 07 November 2017)

The supremum of the set $E_2 = \{1/n : n \in \mathbb{N}\}$ is

Select the correct option

☐	5	<i>AHMAD M-SC(MATHS)</i> <i>GOLD MEDALIST</i> <i>areeb2113@gmail.com</i>
☒	1	
☐	7	
☐	4	

Question # 5 of 10 (Start time: 08:13:37 PM, 07 November 2017)

Identify the false statement(s)

Select the correct option



The set of natural numbers is not complete.

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The set of rational numbers is complete



The set of rational numbers is an infinite set.



The set of real numbers is complete.

Question # 6 of 10 (Start time: 08:14:46 PM, 07 November 2017)

The infimum of set of integers is

Select the correct option



$-\infty$.

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∞ .



-100 .



10 .



Click t

The statement(s) true about the real number $4\sqrt{10}$ is(are)

Select the correct option



is a rational number.

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an irrational number.



the solution of $x^2 - 2x + 3$.



the solution of $x^2 - 4 = 0$.

Which statement(s) is(are) true

Select the correct option



The set of even numbers is defined as $\{2k - 1 : k \in \mathbb{N}\}$.

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The set of odd numbers is defined as $\{2k : k \in \mathbb{N}\}$.



Both set of even numbers and set of odd numbers are countable.



The set of even numbers is a finite set.

Which set is dense in the of real number

Select the correct option



The set of integers.

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The set of rational numbers.



The set $[0, 1]$.



All of the above.

Which one is an irrational number

Select the correct option



$$\frac{22}{7}$$

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$$\pi$$



$$\frac{p+q}{pq}$$
, where p, q are nonnegative integers.



$$\sqrt{s}$$
, where s is a perfect square.

Question # 1 of 10 (Start time: 10:45:18 PM, 16 May 2018)

Which statement is true about the sets $\mathbb{N}$ and $\mathbb{Q}$

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Select the correct option

☐ $\mathbb{N} = \mathbb{Q}$

☐ $\mathbb{Q} \subset \mathbb{N}$.

☒ $\mathbb{N} \subset \mathbb{Q}$.

☐ None of the above.

Click t

Which statement(s) is(are) true

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Select the correct option



The set of even numbers is defined as $\{2k - 1 : k \in \mathbb{N}\}$.



The set of odd numbers is defined as $\{2k : k \in \mathbb{N}\}$.



Both set of even numbers and set of odd numbers are countable.



The set of even numbers is a finite set.

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Select the correct option

☐	5
☒	1
☐	7
☐	4

Click

The supremum of the set $B = \{p \in \mathbb{Q} : p^2 < 3\}$ is

Select the correct option



$\sqrt{2}$.

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$\sqrt{3}$.



10.



19.



Type here to search



The union of set of rational numbers and the set of irrational numbers is

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Select the correct option



The set $\{n \in \mathbb{Q} : x^7 - 10x + 30 = 0\}$.



The set of integers.



The set of natural numbers.



$\mathbb{R}$, the set of real numbers.

The infimum of the set $E_2 = \{1/n : n \in \mathbb{N}\}$ is

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Select the correct option



-1.



2.



-3.



0.

The statement(s) true about the real number $4\sqrt{10}$ is(are)

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Select the correct option



is a rational number.



an irrational number.



the solution of $x^2 - 2x + 3$.



the solution of $x^2 - 4 = 0$.

The intersection of set of odd numbers and the set of even numbers is

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Select the correct option



The set of rational numbers.



The empty set.



$\{1, 2, 3, 4\}$.



The set of real numbers.

Question # 10 of 10 (Start time: 10:49:12 PM, 16 May 2018)

Identify the false statement(s)

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Select the correct option



The set of natural numbers is not complete.



The set of rational numbers is complete



The set of rational numbers is an infinite set.



The set of real numbers is complete.



Which statement(s) is(are) true about the following sequence $s_0 = 1$ and $s_n = 1 - e^{-s_{n-1}}$

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Select the correct option



$0 < s_n \leq 1$ for all n .



$s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}})$.



Sequence is convergent.



All of the above.

$\limsup_{n \rightarrow \infty} s_n = \infty$ if

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Select the correct option



$\{s_n\}$ is not bounded above.



$\{s_n\}$ has unique limit.



$\{s_n\}$ is convergent.



$\{s_n\}$ has limit inferior.



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The series $\sum_{n=0}^{\infty} r^n$, $-1 < r < 1$, is

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Select the correct option



divergent.



convergent and converges to $\frac{1}{1+r^2}$.



convergent and converges to 5.



convergent and converges to $\frac{1}{1-r}$.



The set of terms of $\{s_{n_k}\}$ is contained in the set of terms of $\{s_n\}$ implies

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Select the correct option



$$\sup\{s_n\} = \sup\{s_{n_k}\}.$$



$$\sup\{s_n\} \leq \sup\{s_{n_k}\}.$$



$$\sup\{s_n\} \geq \sup\{s_{n_k}\}.$$



$\{s_n\}$ is convergent.

Every Cauchy sequence has a

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▶ Select the correct option



convergent subsequence.



increasing subsequence.



decreasing subsequence.



positive subsequence.

Question # 10 of 10 (Start time: 02:15:53 PM, 28 November 2017)

The limit of the sequence

$$s_n = \frac{1}{n} + \frac{2(1 + 3/n)}{1 + 1/n}.$$

Select the correct option

☐	1	AHMAD M-SC(MATHS) GOLD MEDALIST areeb2113@gmail.com
☐	4	
☐	3	
☒	2	

If $p > 0$ and α is real, then value of the

$$\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+p)^n},$$

Select the correct option

☐	∞ .	<i>AHMAD M-SC(MATHS)</i> <i>GOLD MEDALIST</i> <i>areeb2113@gmail.com</i>
☐	3.	
☐	2.	
☒	0.	

The value of the

$$\lim_{n \rightarrow \infty} \frac{1}{n^p}$$

Select the correct option

☒	$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$ for $p > 1$.	<i>AHMAD M-SC(MATHS)</i> <i>GOLD MEDALIST</i> <i>areeb2113@gmail.com</i>
☐	$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 3$ for all p .	
☐	$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1$ for all p .	
☐	$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1$ for $p > 1$.	



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Question # 3 of 10 (Start time: 09:47:35 PM, 30 November 2017)

If the sequence is convergent then

Select the correct option



it has two limits.



it is bounded.



it is bounded above but may not be bounded below.



it is bounded below but may not be bounded above.

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Which statement(s) is(are) true about the following sequence $s_0 = 1$ and $s_n = 1 - e^{-s_{n-1}}$

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Select the correct option

☐	$0 < s_n \leq 1$ for all n .
☐	$s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}})$.
☐	Sequence is convergent.
☒	All of the above.



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The series $\sum_{n=0}^{\infty} r^n$, $-1 < r < 1$, is

Select the correct option

☒	divergent.	<i>AHMAD M-SC(MATHS)</i> <i>GOLD MEDALIST</i> <i>areeb2113@gmail-com</i>
☐	convergent and converges to $\frac{1}{1+r^2}$.	
☐	convergent and converges to 5.	
☐	convergent and converges to $\frac{1}{1-r}$.	



Type here to search



Which sequence is bounded

Select the correct option

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$$s_n = [1 + (-1)^n]n.$$



$$\{(-1)^n\}.$$



$$\{n\}.$$



$$\{\frac{1}{n} + n^2\}.$$



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The value of the $\lim_{n \rightarrow \infty} \frac{1}{n^p}$ is

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Select the correct option



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \text{ for } p > 0.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 3 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p > 1.$$



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Every Cauchy sequence has a

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Select the correct option



convergent subsequence.



increasing subsequence.



decreasing subsequence.



positive subsequence.



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Question 17 of 18 (Start time: 00:00:14, 00 November 2017)

Which statement(s) is(are) true for the following sequence $\lim_{n \rightarrow \infty} \sqrt[n]{n}$

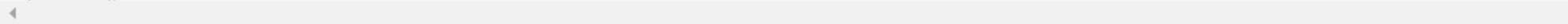
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Select the correct option

☐	The sequence is oscillatory.
☒	The sequence converges to 1.
☐	The sequence converges to 0.
☐	The sequence is divergent to



Which of the following given sets is compact subset of $\mathbb{R}$.

Select the correct option

☐	(1,5]	AHMAD M-SC(MATHS) GOLD MEDALIST areeb2113@gmail.com
☐	{1,2,4}	
☒	[0,1]	
☐	(2,3)	



Type here to search



The supremum of the set $B = \{p \in \mathbb{Q} : p^2 < 3\}$ is

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[Reload Math Equations](#)



$\sqrt{2}$.



$\sqrt{3}$.



10.



19.

[Click to Save Answer & Move to Next Question](#)

Which one is not an initial segment of $\mathbb{N}$

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 Reload Math Equations



$\{1, 2, 3, \dots, 30\}$.



$\{1, 2, 3, \dots, 100, 101, 102, 103\}$.



$\{1, 2, 3\}$.



$\{2, 4, 6\}$.

Click to Save Answer & Move to Next Question

The statement(s) true about the real number $4\sqrt{10}$ is(are)

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 Reload Math Equations



is a rational number.



an irrational number.



the solution of $x^2 - 2x + 3$.



the solution of $x^2 - 4 = 0$.

Click to Save Answer & Move to Next Question

Identify the false statement(s)

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Select the correct option



The set of natural numbers is not complete.



The set of rational numbers is complete



The set of rational numbers is an infinite set.



The set of real numbers is complete.

Click to Save Answer & Move to Next Question

The Infimum of set of integers is:

Select the correct option

☒	$-\infty$
☐	∞
☐	-100
☐	10

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The supremum of a set is

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Select the correct option

- ☐ not unique.
- ☐ the greatest upper bound of that set.
- ☐ least upper bound.
- ☐ None of the above.

The infimum of the set $B = \{p \in \mathbb{Q} : p^2 > 2\}$ is

Select the correct option

☐	$\sqrt{2}$.
☐	$\sqrt{10}$.
☐	$\sqrt{5}$.
☐	3.

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The infimum of the set $B := \{p \in \mathbb{Q} : p^2 > 2\}$ is

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Select the correct option

☐	$\sqrt{2}$
☐	$\sqrt{10}$
☐	$\sqrt{5}$
☐	3

Please do not scroll

The infimum of the set $E_2 = \{1/n : n \in \mathbb{N}\}$ is

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☐	-1.
☐	2.
☐	-3.
☒	0.

Identify the countable set

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Select the correct option

☐	The set of irrational numbers.
☐	The interval $[0, 1]$.
☐	$\mathbb{N} \times \mathbb{N}$.
☐	The interval $[1, 2]$.

Question # 3 of 10 (Start time: 08:47:49 PM, 14 May 2018)

Identify the false statement(s)

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Select the correct option



The set of natural numbers is not complete.



The set of rational numbers is complete



The set of rational numbers is an infinite set.



The set of real numbers is complete.

[Click to Save Answer & Move](#)

The intersection of set of odd numbers and the set of even numbers is

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Select the correct option

☐	The set of rational numbers.
☒	The empty set.
☐	$\{1, 2, 3, 4\}$.
☐	The set of real numbers.



MC170400773: HUMA JAVED

MTH621:Quiz 1

Question # 5 of 10 (Start time: 08:32:36 PM, 14 May 2018)

The union of set of rational numbers and the set of irrational numbers is

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Select the correct option

☐	The set $\{n \in \mathbb{Q} : x^7 - 10x + 30 = 0\}$.
☐	The set of integers.
☐	The set of natural numbers.
☐	$\mathbb{R}$, the set of real numbers.

The supremum of the set $B = \{p \in \mathbb{Q} : p^2 < 3\}$ is

Select the correct option

☐	$\sqrt{2}$.	<i>AHMAD M.SC(MATHS)</i> <i>GOLD MEDALIST</i> <i>areeb2113@gmail.com</i>
☐	$\sqrt{3}$.	
☐	10.	
☐	19.	

Question # 7 of 10 (Start time: 08:34:21 PM, 14 May 2018)

The sum of a rational number and an irrational number is

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Select the correct option

☐	may be a rational depends on the numbers.
☐	always an irrational number.
☐	always a rational number.
☐	may be an irrational number depends on the numbers.

Question # 7 of 10 (Start time: 08:34:21 PM, 14 May 2018)

The sum of a rational number and an irrational number is

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Select the correct option

☐	may be a rational depends on the numbers.
☐	always an irrational number.
☐	always a rational number.
☐	may be an irrational number depends on the numbers.

An upper bound for the set $A = \{p \in \mathbb{Q} : p^2 < 2\}$ is

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Select the correct option

☐ -1

☐ 0

☐ 4

☐ -4

Question # 1 of 10 (Start time: 08:30:27 PM, 14 May 2018)

The supremum of the set $S_2 = \{1/n : n \in \mathbb{N}\}$ is

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Select the correct option

☐	5
☐	1
☐	2
☐	4

ANSWER

Question # 3 of 10 (Start time: 08:31:17 PM, 14 May 2018)

The infimum of the set $E_2 = \{1/n : n \in \mathbb{N}\}$ is*AHMAD M.SC(MATHS)**GOLD MEDALIST**areeb2113@gmail.com*

Select the correct option

☐	-1.
☐	2.
☐	-3.
☒	0.

An upper bound for the set $A = \{p \in \mathbb{Q} : p^2 < 2\}$ is

Select the correct option

☐	-1.	<i>AHMAD M.SC(MATHS)</i> <i>GOLD MEDALIST</i> <i>areeb2113@gmail.com</i>
☐	0.	
☒	4.	
☐	-4.	

Which statement is true about the sets $\mathbb{N}$ and $\mathbb{Q}$ **AHMAD M.SC(MATHS)****GOLD MEDALIST****areeb2113@gmail.com**

Reveal Math Equations

Select the correct option



$\mathbb{N} = \mathbb{Q}$



$\mathbb{Q} \subset \mathbb{N}$



$\mathbb{N} \subset \mathbb{Q}$



None of the above.

This question is compulsory

If the sequence is increasing, then it

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▶ Select the correct option



converges to its supremum.



diverges.



may converges to its supremum.



is bounded.

Which of the following given sets is compact subset of $\mathbb{R}$.

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Select the correct option



(1,5]



{1,2,4}



[0,1]



(2,3)

Identify the identical sequences.

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Select the correct option



$$\left\{ \frac{1}{n^2-2} \right\}_3^\infty \text{ and } \left\{ \frac{1}{n} \right\}_1^\infty.$$



$$\left\{ \frac{1}{n} \right\}_3^\infty \text{ and } \left\{ \frac{1}{n} \right\}_1^\infty.$$



$$\left\{ \frac{1}{n-2} \right\}_3^\infty \text{ and } \left\{ \frac{1}{n^2} \right\}_1^\infty.$$



$$\left\{ \frac{1}{n-2} \right\}_3^\infty \text{ and } \left\{ \frac{1}{n} \right\}_1^\infty.$$